Python for AI Engineering

AI Learning Hub - Comprehensive Course Guide

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Python Fundamentals for AI

Python is the dominant language for AI development.

Key Data Structures:
- Lists: Ordered, mutable sequences. List comprehensions for concise code.
- Dictionaries: Key-value pairs with O(1) average lookup.
- NumPy Arrays: Fixed-type, contiguous memory for fast numerical computation.
- Pandas DataFrames: Tabular data with labeled axes.

NumPy Essentials:
- Broadcasting: Operations on arrays of different shapes.
- Vectorization: Replace loops with array operations for speed.
- Linear algebra: np.dot, np.matmul, np.linalg for matrix operations.
- Random: np.random for reproducible random number generation.

PyTorch and TensorFlow

PyTorch (by Meta) and TensorFlow (by Google) are the two major deep learning frameworks.

PyTorch Basics:
- Tensors: Multi-dimensional arrays with GPU support.
- Autograd: Automatic differentiation for gradient computation.
- nn.Module: Base class for all neural network modules.
- DataLoader: Efficient batched data loading with multiprocessing.
- Training loop: forward pass, loss computation, backward pass, optimizer step.

TensorFlow/Keras:
- tf.keras.Sequential: Stack layers linearly.
- model.compile(): Set optimizer, loss, metrics.
- model.fit(): Train with callbacks (EarlyStopping, ModelCheckpoint).
- TensorBoard: Visualization of training metrics.

MLOps and Deployment

MLOps bridges the gap between ML development and production.

Key Practices:
- Version Control: Git for code, DVC for data and model versioning.
- Experiment Tracking: MLflow, Weights & Biases for logging metrics.
- Model Serving: FastAPI, Flask, TensorFlow Serving, TorchServe.
- Containerization: Docker for consistent environments.
- CI/CD: Automated testing and deployment pipelines.

Model Deployment Options:
- REST APIs: FastAPI endpoint wrapping model.predict().
- Batch Processing: Apache Spark, AWS Batch for large-scale inference.
- Edge Deployment: ONNX Runtime, TensorFlow Lite for mobile/IoT.
- Serverless: AWS Lambda, Google Cloud Functions for auto-scaling.